Jose Luis Costumero Burgos

Independent Researcher

Technical Report v1.1.1 · October 2026 · Open Evaluation Corpus

A Systematic Feasibility Analysis of Smart Contract-Based B2B Services: Lessons from an Adversarial Evaluation of 53 Market Hypotheses and a Future Research Agenda

Jose Luis Costumero Burgos
Independent Researcher
Date: October 2026 · Version: 1.1 (Technical Report — Revised after external feedback)
Repository: `https://github.com/jlcostumero/b2b-smart-contracts-feasibility` (Open Evaluation Corpus)

> ***Author's Note & Disclaimer**: The research, analyses, and conclusions presented in this document were conducted entirely on a personal and independent basis outside of professional employment. The views expressed herein belong strictly to the author and do not reflect or represent the positions, strategies, or opinions of any current or previous employer, affiliated company, or organization.*

Abstract

Academic and industrial literature on Distributed Ledger Technology (DLT) has consistently argued that smart contracts constitute the foundational architecture for a new generation of B2B digital services characterized by disintermediation, tamper-proof evidentiary auditability, and autonomous machine-to-machine (M2M) programmatic settlement. Nevertheless, enterprise adoption of public and Layer-2 (L2) networks remains negligible when compared to established centralized Web2 infrastructures.

In this work, we present a structured adversarial evaluation of **53 commercial hypotheses and business models for smart contract-based B2B services** across EVM-compatible Layer-2 networks (such as Base and Arbitrum One). Each hypothesis was subjected to a sequential five-stage adversarial filtering protocol: European Union regulatory and fiscal compliance (MiCA, DAC8, GDPR, and eIDAS 2.0), technical feasibility and oracle dependency, unit economics against real-time L2 gas costs, enterprise B2B sales and operational friction, and an adversarial *Red-Teaming* counter-analysis against dominant market incumbents.

Our findings reveal an attrition rate of 100% across conventional enterprise domains: none of the 53 hypotheses established an economically defensible competitive moat against established, zero-marginal-cost Web2 alternatives. Drawing from this empirical corpus, we formalize three structural failure heuristics: (1) the *Free Incumbent Heuristic*, demonstrated by the commoditization of cryptographic transparency infrastructures (e.g., Sigstore, GitHub Actions) over commercial on-chain software attestation; (2) the *Legal Identity & AML Friction Vector*, which precludes autonomous M2M settlement under strict European corporate invoicing (CIF/VAT) and KYC/AML mandates; and (3) the *Physical Sensor Oracle Fallacy*. Finally, we outline a **Future Research**

Agenda treating three frontier domains as conditional conjectures: pure agent-to-agent (A2A) coordination without human legal standing, zero-knowledge verifiable SLA enforcement, and censorship-resistant infrastructure.

Keywords—*Smart Contracts, B2B SaaS, Due Diligence, Negative Results, Market Failure, Cyber Resilience Act, MiCA, x402 Protocol, Adversarial Evaluation, Distributed Ledger Technology.*

---

1. Introduction and Problem Formulation

Since Nick Szabo's initial formalization of smart contracts in 1994 and their scalable implementation beginning with Ethereum in 2015, the core theoretical promise of distributed state machines has been the radical reduction of transaction costs by replacing trusted intermediaries with deterministic, self-executing code (*code is law*).

With the maturation of Layer-2 rollups (e.g., Arbitrum One, Base) and protocol-level scalability enhancements such as EIP-4844 (*proto-danksharding*), on-chain data availability costs have decreased by over 90%, reducing standard transaction fees to fractions of a cent ($0.001–$0.01 USD). Simultaneously, the European Union established a comprehensive, predictable legal framework through the entry into force of the Markets in Crypto-Assets Regulation (MiCA, Regulation EU 2023/1114), establishing clear operational boundaries for digital asset services.

Despite these optimal baseline conditions—negligible gas fees, sub-second settlement, and regulatory clarity—enterprise transition toward on-chain computing in non-financial B2B sectors has systematically failed to materialize. The overwhelming majority of enterprise blockchain initiatives have ended in quiet discontinuation. Prominent historical precedents corroborate this pattern across global enterprise consortia: - **TradeLens (Maersk & IBM)**: Developed over five years with estimated cumulative investments exceeding $100M USD to digitize global supply chain documentation, TradeLens announced discontinuation in November 2022 and was fully decommissioned by Q1 2023 due to lack of commercial viability and carrier refusal to join a competitor-governed network [11]. - **we.trade**: A consortium founded by 12 major European banking institutions (including Santander, Deutsche Bank, and HSBC) on Hyperledger Fabric to automate cross-border trade finance via smart contracts, which declared insolvency and appointed liquidators in June 2022 due to insufficient transaction volume [12]. - **Contour & Marco Polo Network**: Major trade finance blockchain consortia (built on R3 Corda) that ceased operations in 2023 following unsustainable operational burn rates and enterprise adoption resistance [13, 14]. - **Australian Stock Exchange (ASX CHESS Replacement)**: Paused in November 2022 following an independent technical review by Accenture [15] that identified severe architectural complexity and batch latency bottlenecks, resulting in a pre-tax derecognition charge of 245–255 million AUD (~170–176 million AUD post-tax). Note that the review attributed failure primarily to distributed application architecture, state synchronization, and project governance rather than DLT consensus limitations alone. - *Architectural Caveat*: These consortia operated on private permissioned networks (Hyperledger Fabric, R3 Corda, Daml); our investigation examines whether public Layer-2 EVM rollups (Base, Arbitrum One) resolve these historical bottlenecks.

These empirical precedents align directly with the foundational decision framework articulated by **Wüst & Gervais (2018)** [10]: when multiple parties share trust in a verification authority, or when an established centralized third party can be held legally accountable under civil jurisdictions, distributed ledgers introduce strictly negative utility in throughput, cost, and operational governance.

This investigation addresses the following core research question:

> *Are there objective economic, technical, or legal conditions under which an L2 smart contract-based B2B service presents a defensible competitive advantage over centralized Web2 architectures within the contemporary European enterprise environment?*

To resolve this question with methodological rigor and counteract the pervasive *LLM optimism and LLM optimism and LLM optimism and LLM optimism and sycophancy bias* (the structural tendency of initial feasibility studies to over-estimate viability), we designed a multi-dimensional adversarial evaluation protocol and applied it across 53 distinct business hypotheses spanning the functional spectrum of modern enterprise operations.

---

2. Multi-Stage Adversarial Evaluation Methodology

The research methodology was structured as a sequential, stage-gated elimination funnel. To be validated as viable for enterprise deployment, a hypothesis was required to pass all five independent evaluation filters without relying on speculative token appreciation, future venture capital subsidies, or regulatory arbitrage.

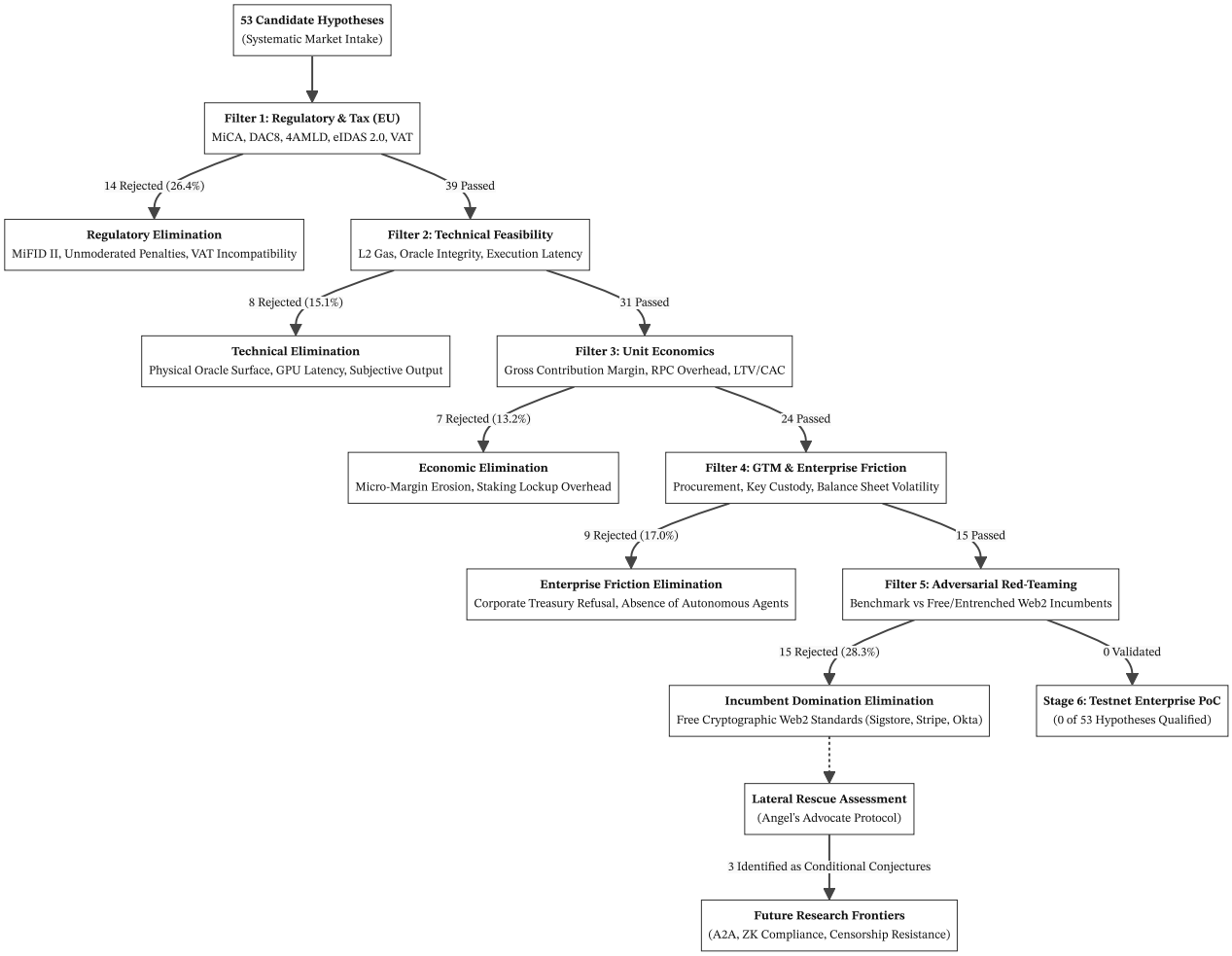


Figure 1: The Multi-Stage Sequential Adversarial Evaluation Funnel for B2B Smart Contract Feasibility.

2.1. Filter 1 · European Regulatory & Tax Compliance

- **MiCA Perimeter (Regulation EU 2023/1114, Art. 3)**: Determination of whether the service qualifies as a Crypto-Asset Service Provider (CASP). Activities requiring formal licensing from national competent authorities (e.g., CNMV, BaFin, ACPR) were audited for compliance overhead.
- **DAC8 Directive & AML/CFT Frameworks**: Assessment under Council Directive (EU) 2023/2226 (DAC8) [3] amending Directive 2011/16/EU on administrative tax cooperation, and Directive (EU) 2015/849 (4AMLD) / Directive (EU) 2018/843 (5AMLD). Verification of mandatory reporting and KYC thresholds when processing digital asset settlements.
- **SaaS Taxation & Corporate Invoicing**: Classification under Directive 2006/112/EC (EU VAT Directive, Art. 58) and national legislation (e.g., Spanish VAT Act 37/1992, Art. 71.3). Evaluation of mandatory invoice formalization with statutory corporate tax numbers (CIF/NIF) and reverse charge mechanics.
- **Evidentiary Legal Standing (eIDAS & eIDAS 2.0)**: Evaluation under Regulation (EU) 910/2014 (eIDAS) and Regulation (EU) 2024/1183 (eIDAS 2.0) [4]. Assessment of electronic timestamping and Qualified Electronic Ledgers (QELs, Arts. 45l–45m). Examination of Art. 41.1 (legal non-discrimination of non-qualified timestamps) versus qualified presumptions of accuracy in judicial proceedings.

2.2. Filter 2 · Technical Feasibility & Infrastructure Overhead

- **L2 Gas Consumption Profiling**: Real-time gas profiling across Base (`chainId: 8453`) and Arbitrum One (`chainId: 42161`). Verification of SSTORE gas consumption (32-byte Merkle root insertion) and calldata costs under live network conditions.
- **Machine Payment Protocols**: Evaluation of compatibility with open programmatic payment standards (HTTP 402 / x402 Protocol [9], 2025 specification), examining `exact` , `upto` , and `batch-settlement` schemes via EIP-3009/Permit2 gasless transfers.
- **Oracle Attack Surface Audit**: Systematic penalization of any service relying on third-party off-chain oracles, which reintroduce the single point of failure and rent-seeking intermediary the architecture aimed to eliminate.

2.3. Filter 3 · Unit Economics & Sustainability

- **Contribution Margin Modeling**: Granular breakdown of gross margins per transaction. Determination of the volume threshold required to cover validator RPC endpoints, failover relays, and infrastructure maintenance without token emissions.
- **Micropayment Viability**: Viability analysis of sub-cent fees ($0.001–$0.01 USD) against payment facilitator settlement overhead and accounting reconciliation costs.

2.4. Filter 4 · Enterprise Go-to-Market & Friction Audit

- **Corporate Buyer Persona Alignment**: Evaluation of procurement friction with enterprise decision-makers (CISOs, CFOs, Heads of Supply Chain).
- **Operational Friction Scoring**: Assessment of private key custody liability, multi-signature corporate governance, and balance-sheet volatility handling under standard IFRS/PGC accounting rules.

2.5. Filter 5 · Adversarial Red-Teaming (Incumbent Stress-Testing)

- **Compulsory Incumbent Confrontation**: Direct benchmark against the most efficient, free, or entrenched centralized alternative (e.g., Sigstore, Stripe, Cloudflare, Okta, Datadog). If an existing Web2 solution resolves ≥90% of the core problem at zero or negligible marginal software cost, the hypothesis is formally rejected due to lack of a defensible moat.

2.6. Scoring Rubric, Veto Gates, and Methodological Boundaries

To ensure systematic comparability, each hypothesis was evaluated across a sequential five-stage elimination funnel utilizing four formal analytical dimensions (scored on a 1.0 to 10.0 scale) and an operational enterprise procurement hurdle at Filter 4: - **Legal & Regulatory Feasibility (L)**: Evaluates strict adherence to EU directives and statutory compliance. Gate: $L < 6.0$ results in immediate rejection at Filter 1. - **Technical Feasibility (T)**: Evaluates implementation viability on EVM L2s without latency or physical oracle dependencies. Gate: $T < 6.0$ results in immediate rejection at Filter 2. - **Economic Viability (E)**: Evaluates positive unit gross margins net of gas, RPC endpoints, and hosting overhead. Gate: $E < 6.0$ results in immediate rejection at Filter 3. - **Enterprise Procurement Friction (Filter 4 Gate)**: Qualitative gate assessing corporate treasury refusal to hold crypto-assets on balance sheets, key custody liability, and the current absence of autonomous agent economic volume. - **Adversarial Robustness (R)**: Measures defensibility against dominant Web2 incumbents. Gate: $R < 6.0$ results in rejection at Filter 5. Across the corpus, R scores cluster around 2.0 to

3.5 (maximum: 3.5 for T-152); because entrenched Web2 platforms provide zero-marginal-cost solutions, incumbent rejection is robust and virtually binary.

Conjunctive Funnel Elimination: The filtering protocol functions as an uncompromising sequential gate: failure in any single filter results in immediate termination of the commercial proposition. Hypotheses failing Filter 1 are disqualified before reaching subsequent filters.

Clarification on Stage 6 (Enterprise Testnet PoC): Pre-implementation screening resulted in 100% attrition across Filters 1 through 5. Consequently, Stage 6 was intentionally not executed: the 0.0% conversion rate (0 / 53) reflects elimination during upstream due diligence, not that 53 testnets were deployed and suffered on-chain crashes.

Methodological Boundaries & Limitations: 1. **Single-Author Qualitative Synthesis**: The evaluation was executed by a single independent researcher. To counteract subjective bias, the analysis incorporated a multi-agent adversarial red-teaming protocol utilizing specialized role personas: *Adversarial Red-Team (Devil's Advocate)*, *EU/Spanish Web3 Legal Auditor*, *Tokenomics & Unit Economics Analyst*, and *Angel's Advocate (Lateral Pivot Search)*. 2. **Permissioned Consortia vs. Public L2s**: Historical enterprise failures (TradeLens, we.trade, Contour, ASX CHESS) operated on permissioned DLTs (Fabric, Corda, Daml). While their failure demonstrated consortium governance fragility, this study focuses on public Layer-2 EVM rollups (Base, Arbitrum One). 3. **Absence of Empirical Stakeholder Interviews**: Conclusions derive from statutory legal texts (EUR-Lex), published enterprise SLA benchmarks, live L2 RPC execution telemetry, and verified corporate API pricing catalogs.

---

3. Hypothesis Taxonomy and Cluster Findings

The 53 evaluated hypotheses were categorized across four primary operational clusters:

Table 1: Reconciled Distribution of Evaluated Hypotheses by Operational Domain.

Functional Cluster	Evaluated Hypotheses	Proportion	Cumulative Rejections
Cluster A: Software Integrity, Artifact Provenance & SBOM	7	13.2%	Filter 5: 5 hypotheses; Filter 1: 2 hypotheses
Cluster B: M2M Settlement, Compute & Autonomous Agent Protocols	16	30.2%	Filter 4: 6 hypotheses; Filter 3: 4; Filter 5: 4; Filter 1: 2
Cluster C: Legal Evidentiary, Notarization & Dispute Resolution	11	20.8%	Filter 1: 6 hypotheses; Filter 5: 3; Filter 2: 2
Cluster D: Supply Chain, Energy & Physical Asset Operations	19	35.8%	Filter 2: 6 hypotheses; Filter 1: 4; Filter 5: 3; Filter 3: 3; Filter 4: 3
Total Evaluated Corpus	**53**	**100.0%**	**Stage 6 Conversion Rate: 0.0% (0 / 53)**

Cluster A: Software Integrity, Artifact Provenance & SBOM (7 Hypotheses)

Focusing on cryptographic build provenance, container signing, and Software Bill of Materials (SBOM) anchoring to address the **Cyber Resilience Act (CRA, Regulation EU 2024/2847)** and NIS2 directives (`T-`

`134`, `T-137`, `T-140`, `T-141`, `T-143`, `T-147`, `T-150`).

- **Initial Evaluation**: Recorded the highest baseline technical scores across the study (Mean: **8.0 / 10** for `T-140`; Legal: 8.0, Technical: 8.5, Economic: 7.5).
- **The Rejection Point**: Overwhelming incumbent bundling by **Sigstore, GitHub Actions, and Chainguard**. For public open-source software, GitHub provides free SLSA Level 2 [8] build provenance anchored to the Rekor transparency log (operated by Google and the Linux Foundation) at zero marginal cost. For enterprise private repositories, while GitHub Enterprise Cloud utilizes private Sigstore/cosign instances rather than the public Rekor log, this functionality is bundled directly into existing corporate seat licenses. Enterprise CISOs will not license a third-party smart contract startup when their primary CI/CD platform already generates free, auditor-recognized cryptographically verifiable provenance.

Cluster B: M2M Settlement, Compute & Autonomous Agent Protocols (16 Hypotheses)

Focusing on pay-per-call API access, autonomous agent compute billing, and micro-escrows using the HTTP 402 / x402 [9] standard settled in USDC on Base (`T-101`, `T-104`, `T-118`, `T-126`, `T-127`, `T-128`, `T-129`, `T-130`, `T-132`, `T-135`, `T-136`, `T-138`, `T-139`, `T-144`, `T-148`, `T-152`).

- **Demonstrated Strengths**: Impeccable technical feasibility. The open x402 standard enables programmatic machine settlement without user account creation or manual credit card entry.
- **The Rejection Point**: Absence of commercial demand mass and corporate treasury friction. In 2026, the autonomous agent economy operating self-funded on-chain treasury remains an emerging frontier. For human software engineers and enterprise platforms, established billing APIs (Stripe Billing, AWS Marketplace) eliminate accounting friction, tax reconciliation complexities, and legal ambiguity. In FX settlement (`T-118`), corporate B2B SME settlement in Europe is efficiently captured by SEPA Instant, low-cost multi-currency accounts (Wise Business, Revolut Business with 0.1–0.4% spreads), and automated bank APIs, where currency risk is zero compared to stablecoin on/off-ramp spreads (0.3%–1.5%) plus tax reporting liability.

Cluster C: Legal Evidentiary, Notarization & Dispute Resolution (11 Hypotheses)

Encompassing decentralized timestamping (`T-102`), binding arbitration (`T-106`), penalty enforcement (`T-110`), ESG hash compliance (`T-112`), SLA monitoring (`T-113`), consortium governance (`T-115`), IP licensing (`T-123`), public procurement (`T-142`), Tor ZK timestamping (`T-145`), R&D IP escrow (`T-151`), and TDM copyright remuneration (`T-153`).

- **The Rejection Point**: European Union evidentiary law and judicial reality. Under Regulation (EU) 910/2014 (eIDAS) and Regulation (EU) 2024/1183 (eIDAS 2.0) [4], Articles 45l and 45m formally introduce *Qualified Electronic Ledgers* and grant data recorded in them a legal presumption of unique and accurate sequential chronological ordering and integrity. However, public Layer-2 networks (Base, Arbitrum One) do not currently possess Qualified Trust Service Provider (QTSP) designation under Art. 45m; thus, public on-chain transaction hashes operate as non-qualified electronic ledgers (Art. 45l(1)), enjoying basic judicial admissibility but lacking the statutory qualified presumption of accuracy that established RFC 3161 QTSPs provide free or for pennies (FreeTSA, BaseStamp). Furthermore, automated, unstoppable contractual penalty enforcement (`T-110`) conflicts directly with statutory civil law doctrines (such as Spanish Civil Code Art. 1154), which empower judges to equitably moderate penalties when obligations are partially or irregularly fulfilled.

Cluster D: Supply Chain, Energy & Physical Asset Operations (19 Hypotheses)

Addressing regulatory certificates (`T-103`), machinery telemetry (`T-105`), maintenance settlement (`T-107`), equipment leases (`T-108`), supply chain provenance (`T-109`), parametric insurance (`T-111`), SaaS licensing (`T-114`), IAM credentials (`T-116`), revenue splits (`T-117`), freight delivery payments (`T-119`), PPA energy settlement (`T-120`), letters of credit (`T-121`), M&A earnouts (`T-122`), payroll (`T-124`), carbon credits (`T-125`), agent reputation (`T-131`), inference slashing (`T-133`), cold-chain pharma tracking (`T-146`), and industrial ESG telemetry (`T-149`).

- **The Rejection Point**: The Physical Sensor Oracle Fallacy and entrenched institutional monopolies. A smart contract guarantees that data, once anchored, cannot be altered. However, it **cannot verify that the physical sensor taking the measurement was accurate, calibrated, or free from hardware spoofing at the point of capture**. Enterprise compliance departments rely on certified closed hardware (SCADA systems, Siemens/Rockwell controllers with hardware TPM/HSM enclaves) and audited relational databases. In energy (`T-120`) and carbon credits (`T-125`), national grid operators (REE/OMIE) and sovereign registries (EU ETS) operate legal monopolies that reject third-party on-chain settlement.

---

4. Causal Analysis: The Three Structural Failure Heuristics

Cross-sectional synthesis of the 53 hypotheses reveals three structural economic and technical heuristics that systematically undermine smart contract-based B2B offerings:

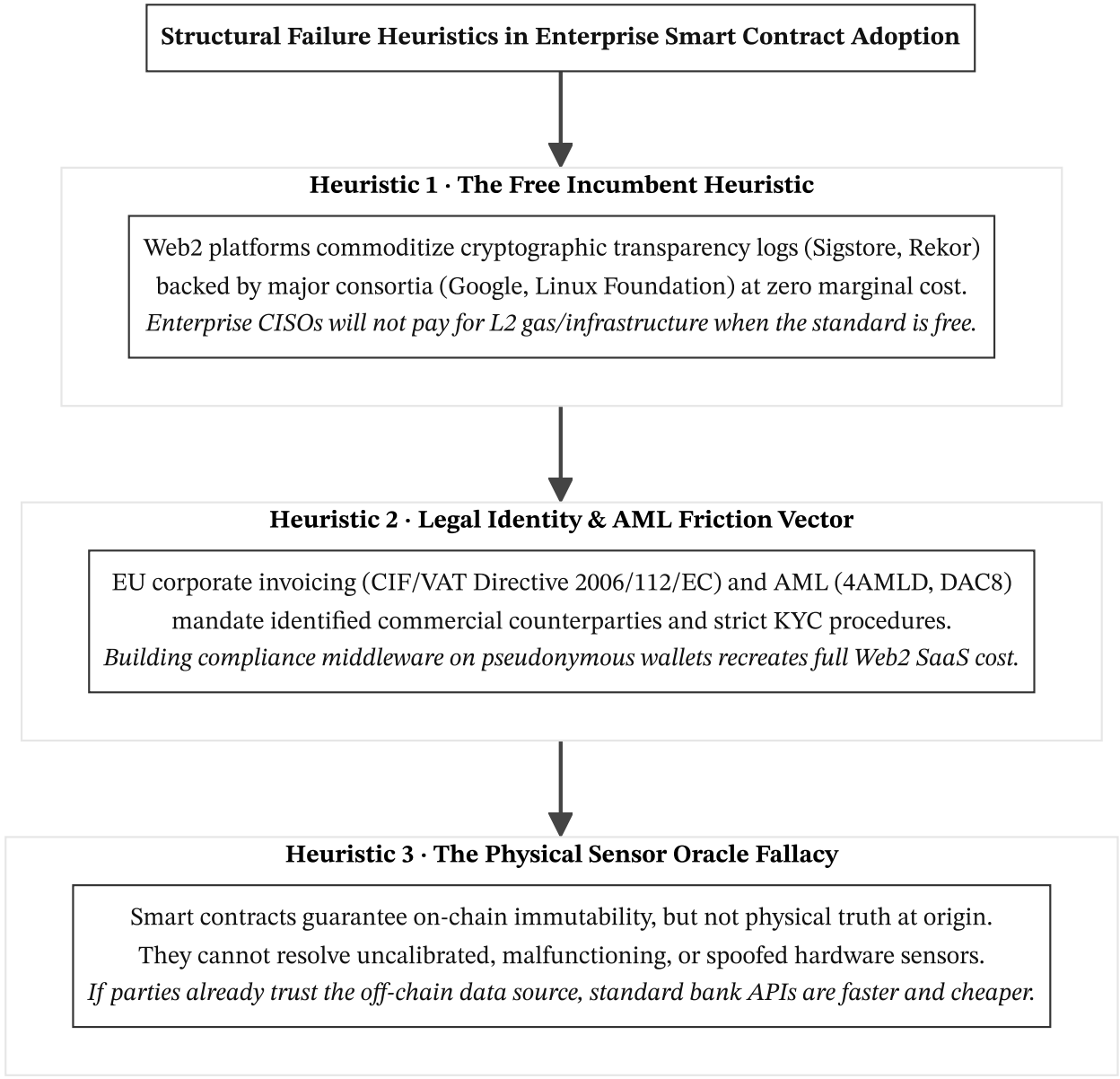


Figure 2: The Three Structural Failure Heuristics in Enterprise Smart Contract Adoption.

4.1. The Free Incumbent Heuristic (Commoditized Trust)

In **37 of the 53 evaluated cases (69.8%)**, the proposed value proposition centered on *tamper-proof immutability and vendor independence*. However, the modern Web2 engineering ecosystem has already commodified distributed cryptographic primitives (Merkle trees, OIDC keyless signing, RFC 6962 Certificate Transparency logs) and embedded them into existing workflows at zero marginal cost.

When major industry consortia (Google, Microsoft, Cloudflare, Linux Foundation) operate public, auditable append-only ledgers, corporate buyers consistently favor the institutional reliability, legal backing, and zero cost of those platforms over the theoretical decentralization of an L2 network.

4.2. The Legal Identity & Tax Compliance Friction Vector

Smart contracts operate with mathematical precision between pseudonymous public keys (`0x...`). However, corporate B2B commerce in the European Union mandates: 1. Formal commercial invoices containing statutory corporate tax identifiers (CIF/NIF), registered company names, and applicable VAT rates (Directive 2006/112/EC). 2. Adherence to tax transparency and anti-money laundering regulations: Council Directive (EU) 2023/2226 (DAC8) [3] mandates automatic reporting and exchange of crypto-asset transaction data with tax authorities, while Directive (EU) 2015/849 (4AMLD), Regulation (EU) 2024/1624 (AMLR) [17], and Regulation (EU) 2023/1113 (Travel Rule Regulation) [18] mandate full originator/beneficiary identification, making anonymous or unhosted wallet B2B transfers non-compliant. 3. Accounting adherence to IFRS/national accounting standards regarding digital asset holdings and foreign exchange gains/losses.

Any platform attempting to enable real-time on-chain B2B micropayments must construct comprehensive traditional invoicing, tax withholding, and identity-verification middleware. This requirement recreates the exact cost structure and overhead of an ordinary Web2 SaaS provider, extinguishing the native efficiency advantage of the blockchain.

4.3. The Physical Sensor Oracle Fallacy (Garbage In, Garbage Out)

Numerous hypotheses attempted to utilize smart contracts as automated arbiters of commercial penalties or parametric insurance payouts based on physical world triggers (e.g., shipping delays, cold-chain temperature violations, SLA downtimes).

In all instances, the smart contract is structurally redundant: if both commercial parties already agree to trust the off-chain data source feeding the contract (e.g., a Maersk container GPS or a validated pharmaceutical logger), executing settlement via a standard bank API or credit line is faster, cheaper, and avoids the financial opportunity cost of immobilizing liquidity in smart contract escrows.

---

5. Future Research Agenda: Conjectures and Conditional Utility

The systematic rejection of conventional B2B blockchain applications does not indicate that distributed state machines possess zero utility. Rather, it **delineates with precision where their defensible utility begins**: exclusively when **centralized Web2 architectures are architecturally, politically, or legally incapable of functioning**.

We propose three conditional frontier research directions, formulating explicit **falsifiability criteria** for each:

Table 2: Priority Frontier Research Directions and Falsifiability Criteria.

Frontier Direction	Primary Technical Enabler	Non-Substitutable Utility	Falsifiability Criterion (Conditions to Validate)
1. Autonomous Agent Coordination (A2A)	ERC-8004 + HTTP 402 (x402 Protocol)	Micro-settlement for autonomous AI models lacking human legal identity, bank accounts, or corporate registration.	Sustained operational deployment of swarms controlling > $100k treasury executing > 10,000 monthly tx without human fiat rails.

Frontier Direction	Primary Technical Enabler	Non-Substitutable Utility	Falsifiability Criterion (Conditions to Validate)
2. Zero-Knowledge Verifiable Compliance	ZK-SNARKs (ZK-SLA & Private Provenance)	Proof of regulatory (CRA) or SLA compliance without exposing source code, trade secrets, or dependency graphs.	Formal adoption by EU regulatory bodies or defense procurement agencies accepting ZK proofs in lieu of source code escrow.
3. Censorship-Resistant Infrastructure	Tor Onion Routing + L2 Rollups	Unstoppable settlement and compute coordination across jurisdictions where traditional payment rails refuse or block service.	Continuous operation of independent compute or research nodes under active jurisdictional de-platforming.

Direction 1: Autonomous Agent-to-Agent Coordination (ERC-8004 + x402)

- **Context**: Human developers and registered corporations will always prefer SEPA transfers, Stripe, or corporate credit lines due to statutory tax and accounting convenience. However, **autonomous AI agents** (local LLM instances, autonomous swarms) possess no legal identity, cannot hold traditional bank accounts, and cannot execute corporate contracts.
- **Conjecture**: Permissionless machine-to-machine payment rails leveraging **ERC-8004** agent identity standards and **HTTP 402 (x402)** protocols enable AI agents to purchase compute, storage, and contextual inference programmatically via stablecoins without human intermediation.
- **Falsifiability / Stress-Test**: This conjecture is falsified if centralized agent payment protocols (e.g., Visa/Stripe agentic payment protocols or Google AP2) provide sub-agent virtual cards with automated KYC linked to the human operator, eliminating the need for on-chain wallets.

Direction 2: Zero-Knowledge Verifiable Compliance (ZK-SLA & Private Provenance)

- **Context**: In multi-party industrial consortia and defense aerospace supply chains, enterprises strictly refuse to publish build manifests or telemetry to public transparency logs (like Sigstore/Rekor [7]) because doing so leaks proprietary dependencies and trade secrets.
- **Conjecture**: Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge (ZK-SNARKs) can mathematically prove compliance with regulations (such as the EU Cyber Resilience Act) or contractual SLAs without revealing underlying source code, contributor identities, or network architecture.
- **Falsifiability / Stress-Test**: This conjecture is falsified if confidential computing enclaves (Intel TDX, AMD SEV-SNP) and trusted audit clearinghouses deliver cryptographic verification at 1/100th the compute proving cost of ZK-SNARKs.

Direction 3: Censorship-Resistant Operational Infrastructure

- **Context**: Centralized Web2 payment processors (Stripe, PayPal) and hosting providers (AWS, Cloudflare) operate under strict territorial jurisdictional constraints, subject to de-platforming, political sanctions, and unilateral account freezes.
- **Structural Regulatory Tension**: This direction operates in direct, fundamental tension with Filter 1 compliance (EU AML Single Rulebook, MiCA, and Travel Rule Regulation 2023/1113). It is viable exclusively in pure machine-to-machine sub-economies operating outside statutory legal frameworks, not for registered tax-paying corporations.

- **Conjecture**: Cryptographic payment routing protocols operating over privacy-preserving transport layers (Tor onion routing, I2P) combined with L2 zero-knowledge rollups provide resilient micro-settlement for whistleblowing, independent security research, and censorship-resistant communications.
- **Falsifiability / Stress-Test**: This conjecture is falsified if network latency (> 2,500ms on multi-hop Tor circuits) and liquidity fragmentation make automated agent interaction unviable compared to decentralized VPNs (WireGuard overlays).

---

6. Conclusions

The systematic evaluation of 53 market hypotheses demonstrates that attempting to migrate conventional B2B enterprise workflows onto smart contracts represents a fundamental architectural category error. In the overwhelming majority of commercial applications, centralized authority is not an inefficiency to be eliminated, but an intentional, cost-effective feature that provides legal clarity, sub-millisecond execution, judicial recourse, and zero marginal software cost.

The true value of an adversarial due diligence methodology is not to manufacture false optimism, but to **prevent the catastrophic misallocation of engineering talent and capital toward solutions searching for a problem**. The future of smart contract technology will not replace human enterprise Web2 software; rather, it will serve as the native financial and evidentiary coordination substrate for the emerging economy of autonomous machine agents operating beyond the boundaries of traditional banking.

---

7. References and Regulatory Sources

[1] **European Parliament & Council** (2023). *Regulation (EU) 2023/1114 on markets in crypto-assets (MiCA)*. Official Journal of the European Union, L 150/40.
[2] **European Parliament & Council** (2024). *Regulation (EU) 2024/2847 on horizontal cybersecurity requirements for products with digital elements (Cyber Resilience Act - CRA)*. Official Journal of the European Union, L 2024/2847.
[3] **Council of the European Union** (2023). *Council Directive (EU) 2023/2226 amending Directive 2011/16/EU on administrative cooperation in the field of taxation (DAC8)*. Official Journal of the European Union, L 2023/2226.
[4] **European Parliament & Council** (2024). *Regulation (EU) 2024/1183 amending Regulation (EU) No 910/2014 as regards establishing the European Digital Identity Framework (eIDAS 2.0 and Qualified Electronic Ledgers)*. Official Journal of the European Union, L 2024/1183.
[5] **European Parliament & Council** (2015). *Directive (EU) 2015/849 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing (4AMLD)*. Official Journal of the European Union, L 141/73.
[6] **Spanish Parliament** (1992). *Spanish Law 37/1992 on Value Added Tax (VAT Act)*, Art. 71.3 (Electronically supplied services). BOE-A-1992-28740.
[7] **Sigstore Project** (2024). *Sigstore: Software Signing and Transparency Architecture*. OpenSSF / Linux Foundation. `https://docs.sigstore.dev`.
[8] **OpenSSF** (2023). *Supply-chain Levels for Software Artifacts (SLSA) Specification v1.0*. `https://slsa.dev/spec/v1.0/`.

[9] **Coinbase Developer Platform** (2025). *x402: An Open Protocol for Machine-to-Machine Payments on Base*. `https://github.com/coinbase/x402`.

[10] **Wüst, K., & Gervais, A.** (2018). *Do you need a blockchain?* In 2018 Crypto Valley Conference on Blockchain Technology (CVCBT) (pp. 45-54). IEEE.

[11] **A.P. Moller - Maersk & IBM** (2022). *TradeLens to be discontinued*. Press Release, November 29, 2022. Decommissioned Q1 2023.

[12] **Global Trade Review** (2022). *Bank-backed blockchain consortium we.trade halts operations and enters liquidation*. GTR News, June 2022.

[13] **Global Trade Review** (2023). *Contour trade finance network shuts down*. GTR News, November 2023.

[14] **Ledger Insights** (2023). *Trade finance consortium Marco Polo Network files for insolvency*. News Report, February 2023.

[15] **Accenture** (2022). *Independent Technical Review of the ASX CHESS Replacement Project*. Australian Securities Exchange Technical Assessment Report, November 2022.

[16] **Chainguard** (2024). *Enterprise Software Bill of Materials (SBOM) and Continuous Artifact Attestation*. Technical Whitepaper.

[17] **Council of the European Union** (2006). *Council Directive 2006/112/EC of 28 November 2006 on the common system of value added tax*. Official Journal of the European Union, L 347/1.

[18] **European Parliament & Council** (2024). *Regulation (EU) 2024/1624 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing (AMLR)*. Official Journal of the European Union, L 2024/1624.

[19] **European Parliament & Council** (2023). *Regulation (EU) 2023/1113 on information accompanying transfers of funds and certain crypto-assets (Travel Rule)*. Official Journal of the European Union, L 156/1.

---

8. Appendix: Master Matrix of 53 Evaluated Hypotheses

Table 3 (Master Evaluation Matrix): Comprehensive Catalogue of 53 Evaluated B2B Smart Contract Hypotheses.

The following table catalogs all 53 research hypotheses evaluated during this study, detailing functional identifier, English denomination, thematic cluster, elimination stage, multi-dimensional scores (**L**: Legal, **T**: Technical, **E**: Economic, **R**: Adversarial Robustness), dominant incumbent alternative, and the critical failure factor:

ID	Hypothesis Denomination	Thematic Cluster	Elimination Stage	Scores (L/T/E/R)	Dominant Incumbent Alternative	Critical Rejection Factor
T-101	**AI Intelligence API Gateway with Pay-per-Call (x402)**	M2M Settlement & Compute	**Filter 4 (Procurement)**	L:6.0 T:8.0 E:7.5	L:6.0 T:8.0 E:7.5 R:3.0	L:6.0 T:8.0 E:7.5 R:3.0
T-102	**Decentralized B2B Timestamping and Notarization Protocol**	Legal Evidentiary & Notarization	**Filter 5 (Incumbent)**	L:6.0 T:8.0 E:8.0	L:6.0 T:8.0 E:8.0 R:2.0	L:6.0 T:8.0 E:8.0 R:2.0
T-103	**Automated Regulatory Compliance Certificate Verification**	Supply Chain & Physical Assets	**Filter 4 (Procurement)**	L:6.0 T:7.0 E:7.5	L:6.0 T:7.0 E:7.5 R:2.0	L:6.0 T:7.0 E:7.5 R:2.0
T-104	**Programmatic B2B Contract Settlement with Verified Milestones**	M2M Settlement & Compute	**Filter 4 (Procurement)**	L:7.0 T:8.0 E:7.0	L:7.0 T:8.0 E:7.0 R:2.0	L:7.0 T:8.0 E:7.0 R:2.0
T-105	**Industrial Sensor Data Anchoring with Multi-Oracle Verification**	Supply Chain & Physical Assets	**Filter 2 (Technical)**	L:6.0 T:5.5 E:6.0	L:6.0 T:5.5 E:6.0 R:3.0	L:6.0 T:5.5 E:6.0 R:3.0
T-106	**On-Chain Binding Arbitration Protocol for Commercial Disputes**	Legal Evidentiary & Notarization	**Filter 1 (Regulatory)**	L:4.0 T:7.0 E:5.5	L:4.0 T:7.0 E:5.5 R:2.0	L:4.0 T:7.0 E:5.5 R:2.0
T-107	**Automated Industrial Maintenance Contracts with Settlement**	Supply Chain & Physical Assets	**Filter 3 (Economics)**	L:6.0 T:7.0 E:5.5	L:6.0 T:7.0 E:5.5 R:3.0	L:6.0 T:7.0 E:5.5 R:3.0
T-108	**Industrial Equipment Lease Cash Flow Tokenization (RWA)**	Supply Chain & Physical Assets	**Filter 1 (Regulatory)**	L:4.5 T:6.0 E:6.0	L:4.5 T:6.0 E:6.0 R:2.0	L:4.5 T:6.0 E:6.0 R:2.0
T-109	**Supply Chain Provenance with On-Chain Evidence Settlement**	Supply Chain & Physical Assets	**Filter 5 (Incumbent)**	L:6.0 T:8.0 E:6.5	L:6.0 T:8.0 E:6.5 R:2.0	L:6.0 T:8.0 E:6.5 R:2.0
T-110	**Programmatic Contractual Penalty Enforcement Protocol**	Legal Evidentiary & Notarization	**Filter 1 (Regulatory)**	L:3.5 T:7.0 E:6.5	L:3.5 T:7.0 E:6.5 R:3.0	L:3.5 T:7.0 E:6.5 R:3.0
T-111	**Parametric Insurance Trigger Monitoring and Settlement**	Supply Chain & Physical Assets	**Filter 3 (Economics)**	L:6.0 T:7.0 E:5.5	L:6.0 T:7.0 E:5.5 R:2.0	L:6.0 T:7.0 E:5.5 R:2.0
T-112	**Programmatic ESG Compliance Management Protocol**	Legal Evidentiary & Notarization	**Filter 2 (Technical)**	L:6.0 T:5.5 E:6.0	L:6.0 T:5.5 E:6.0 R:2.5	L:6.0 T:5.5 E:6.0 R:2.5
T-113	**Digital Service SLA Monitoring with Evidentiary Records**	Legal Evidentiary & Notarization	**Filter 5 (Incumbent)**	L:6.0 T:7.0 E:7.5	L:6.0 T:7.0 E:7.5 R:2.0	L:6.0 T:7.0 E:7.5 R:2.0
T-114	**Multi-Tenant SaaS Permission and License Management Protocol**	Supply Chain & Physical Assets	**Filter 5 (Incumbent)**	L:6.5 T:7.0 E:6.5	L:6.5 T:7.0 E:6.5 R:2.0	L:6.5 T:7.0 E:6.5 R:2.0
T-115	**Consortium Industrial Governance Protocol with Staking**	Legal Evidentiary & Notarization	**Filter 1 (Regulatory)**	L:4.0 T:7.0 E:4.0	L:4.0 T:7.0 E:4.0 R:2.0	L:4.0 T:7.0 E:4.0 R:2.0

ID	Hypothesis Denomination	Thematic Cluster	Elimination Stage	Scores (L/T/E/R)	Dominant Incumbent Alternative	Critical Rejection Factor
T-116	Multi-Tenant IAM Protocol with On-Chain Credentials	Supply Chain & Physical Assets	Filter 2 (Technical)	L:6.0 T:5.0 E:6.5 R:2.0	L:6.0 T:5.0 E:6.5	L:6.0 T:5.0 E:6.5 R:2.0
T-117	Automated Revenue Split-as-a-Service Protocol	Supply Chain & Physical Assets	Filter 1 (Regulatory)	L:4.5 T:8.0 E:8.5 R:2.5	L:4.5 T:8.0 E:8.5	L:4.5 T:8.0 E:8.5 R:2.5
T-118	FX Settlement-as-a-Service Protocol for Cross-Border B2B	M2M Settlement & Compute	Filter 1 (Regulatory)	L:5.0 T:7.0 E:8.0 R:3.0	L:5.0 T:7.0 E:8.0	L:5.0 T:7.0 E:8.0 R:3.0
T-119	Delivery-Triggered Payment-as-a-Service Protocol	Supply Chain & Physical Assets	Filter 4 (Procurement)	L:6.0 T:7.0 E:7.5 R:3.0	L:6.0 T:7.0 E:7.5	L:6.0 T:7.0 E:7.5 R:3.0
T-120	Programmatic Settlement of Power Purchase Agreements (PPA)	Supply Chain & Physical Assets	Filter 1 (Regulatory)	L:4.0 T:7.0 E:7.0 R:2.0	L:4.0 T:7.0 E:7.0	L:4.0 T:7.0 E:7.0 R:2.0
T-121	Cross-Border Trade Finance: Automated Letter of Credit	Supply Chain & Physical Assets	Filter 2 (Technical)	L:6.0 T:5.5 E:6.0 R:2.0	L:6.0 T:5.5 E:6.0	L:6.0 T:5.5 E:6.0 R:2.0
T-122	Cross-Border M&A: Programmatic Earnout Escrow Release	Supply Chain & Physical Assets	Filter 2 (Technical)	L:6.0 T:5.5 E:6.0 R:2.0	L:6.0 T:5.5 E:6.0	L:6.0 T:5.5 E:6.0 R:2.0
T-123	Cross-Border IP Licensing: Royalty Distribution Protocol	Legal Evidentiary & Notarization	Filter 1 (Regulatory)	L:4.5 T:6.0 E:6.0 R:2.0	L:4.5 T:6.0 E:6.0	L:4.5 T:6.0 E:6.0 R:2.0
T-124	Cross-Border Payroll Settlement Protocol for Remote Teams	Supply Chain & Physical Assets	Filter 1 (Regulatory)	L:4.0 T:6.0 E:6.5 R:3.0	L:4.0 T:6.0 E:6.5	L:4.0 T:6.0 E:6.5 R:3.0
T-125	Cross-Border Carbon Credit Settlement and Retirement	Supply Chain & Physical Assets	Filter 2 (Technical)	L:6.0 T:5.5 E:6.0 R:2.0	L:6.0 T:5.5 E:6.0	L:6.0 T:5.5 E:6.0 R:2.0
T-126	Agent-to-Agent Micro-Arbitrage of L2 DEX Liquidity	M2M Settlement & Compute	Filter 5 (Incumbent)	L:6.0 T:7.0 E:6.5 R:2.0	L:6.0 T:7.0 E:6.5	L:6.0 T:7.0 E:6.5 R:2.0
T-127	Agent-to-Agent Settlement with ERC-4337 and Escrows	M2M Settlement & Compute	Filter 4 (Procurement)	L:6.5 T:7.0 E:6.5 R:3.0	L:6.5 T:7.0 E:6.5	L:6.5 T:7.0 E:6.5 R:3.0
T-128	Autonomous Agent Orchestrator with Collateral Staking	M2M Settlement & Compute	Filter 4 (Procurement)	L:6.0 T:7.0 E:6.5 R:2.0	L:6.0 T:7.0 E:6.5	L:6.0 T:7.0 E:6.5 R:2.0
T-129	Agentic Market Maker for Private Cloud Computing Services	M2M Settlement & Compute	Filter 3 (Economics)	L:6.0 T:7.0 E:4.0 R:2.0	L:6.0 T:7.0 E:4.0	L:6.0 T:7.0 E:4.0 R:2.0
T-130	Cryptographically Conditioned Escrow for AI Negotiators	M2M Settlement & Compute	Filter 5 (Incumbent)	L:6.0 T:7.0 E:6.5 R:2.0	L:6.0 T:7.0 E:6.5	L:6.0 T:7.0 E:6.5 R:2.0
T-131	Decentralized Reputation Registry for AI Agents	Supply Chain & Physical Assets	Filter 3 (Economics)	L:6.0 T:7.0 E:5.5 R:2.0	L:6.0 T:7.0 E:5.5	L:6.0 T:7.0 E:5.5 R:2.0
T-132	Multi-Agent Coordination Protocol with Pay-per-Inference	M2M Settlement & Compute	Filter 3 (Economics)	L:6.0 T:6.0 E:5.5 R:2.0	L:6.0 T:6.0 E:5.5	L:6.0 T:6.0 E:5.5 R:2.0
T-133	AI Inference Quality Verification Protocol with Slashing	Supply Chain & Physical Assets	Filter 2 (Technical)	L:6.0 T:5.5 E:5.5 R:2.0	L:6.0 T:5.5 E:5.5	L:6.0 T:5.5 E:5.5 R:2.0

ID	Hypothesis Denomination	Thematic Cluster	Elimination Stage	Scores (L/T/E/R)	Dominant Incumbent Alternative	Critical Rejection Factor
T-134	AI Model Training Data Marketplace and Settlement Protocol	Software Integrity & Provenance	Filter 1 (Regulatory)	L:5.0 T:7.0 E:7.0 R:2.0	L:5.0 T:7.0 E:7.0	L:5.0 T:7.0 E:7.0 R:2.0
T-135	Distributed GPU Compute Pool with Proof-of-Work Settlement	M2M Settlement & Compute	Filter 3 (Economics)	L:6.0 T:7.0 E:5.5 R:2.0	L:6.0 T:7.0 E:5.5	L:6.0 T:7.0 E:5.5 R:2.0
T-136	x402 Compute Utility for ERC-8004 Autonomous Agents	M2M Settlement & Compute	Filter 4 (Procurement)	L:7.0 T:7.5 E:7.5 R:3.0	L:7.0 T:7.5 E:7.5	L:7.0 T:7.5 E:7.5 R:3.0
T-137	On-Chain AI Model Weights Verification and Watermarking	Software Integrity & Provenance	Filter 5 (Incumbent)	L:6.5 T:8.0 E:6.5 R:2.0	L:6.5 T:8.0 E:6.5	L:6.5 T:8.0 E:6.5 R:2.0
T-138	Private Inference Server on Tor Network with Escrow	M2M Settlement & Compute	Filter 5 (Incumbent)	L:6.0 T:6.5 E:6.5 R:2.0	L:6.0 T:6.5 E:6.5	L:6.0 T:6.5 E:6.5 R:2.0
T-139	GPU Hash-Rate Derivatives Protocol (Cloud Compute)	M2M Settlement & Compute	Filter 1 (Regulatory)	L:4.0 T:7.5 E:6.5 R:2.0	L:4.0 T:7.5 E:6.5	L:4.0 T:7.5 E:6.5 R:2.0
T-140	Cryptographic Build Integrity Attestation (SLSA Level 2)	Software Integrity & Provenance	Filter 5 (Incumbent)	L:8.0 T:8.5 E:7.5 R:2.0	L:8.0 T:8.5 E:7.5	L:8.0 T:8.5 E:7.5 R:2.0
T-141	Open Data Metered Licensing Protocol with Settlement	Software Integrity & Provenance	Filter 5 (Incumbent)	L:6.5 T:7.5 E:6.5 R:2.0	L:6.5 T:7.5 E:6.5	L:6.5 T:7.5 E:6.5 R:2.0
T-142	Programmatic Bid Verification for Public Procurement	Legal Evidentiary & Notarization	Filter 1 (Regulatory)	L:4.5 T:6.0 E:6.5 R:2.0	L:4.5 T:6.0 E:6.5	L:4.5 T:6.0 E:6.5 R:2.0
T-143	On-Chain SBOM Attestation for Cyber Resilience Act	Software Integrity & Provenance	Filter 5 (Incumbent)	L:6.5 T:8.5 E:7.0 R:2.0	L:6.5 T:8.5 E:7.0	L:6.5 T:8.5 E:7.0 R:2.0
T-144	P2P Payment Routing Protocol for Censorship-Resistant Agents	M2M Settlement & Compute	Filter 3 (Economics)	L:6.0 T:6.5 E:5.5 R:2.0	L:6.0 T:6.5 E:5.5	L:6.0 T:6.5 E:5.5 R:2.0
T-145	Sovereign Tor/P2P Timestamping Protocol with ZK Proofs	Legal Evidentiary & Notarization	Filter 2 (Technical)	L:6.0 T:5.5 E:7.0 R:2.5	L:6.0 T:5.5 E:7.0	L:6.0 T:5.5 E:7.0 R:2.5
T-146	Cold Chain Integrity Attestation Protocol (Pharma Traceability)	Supply Chain & Physical Assets	Filter 4 (Procurement)	L:6.0 T:8.0 E:7.5 R:2.0	L:6.0 T:8.0 E:7.5	L:6.0 T:8.0 E:7.5 R:2.0
T-147	Proof-of-Training: Provenance for Frontier AI Models	Software Integrity & Provenance	Filter 5 (Incumbent)	L:6.5 T:7.5 E:6.0 R:2.0	L:6.5 T:7.5 E:6.0	L:6.5 T:7.5 E:6.0 R:2.0
T-148	P2P Compute Orchestration on Hidden Networks with ZK	M2M Settlement & Compute	Filter 5 (Incumbent)	L:6.0 T:6.0 E:6.5 R:2.0	L:6.0 T:6.0 E:6.5	L:6.0 T:6.0 E:6.5 R:2.0
T-149	Cryptographic Industrial Telemetry for ESG Compliance	Supply Chain & Physical Assets	Filter 5 (Incumbent)	L:6.0 T:7.5 E:7.0 R:2.0	L:6.0 T:7.5 E:7.0	L:6.0 T:7.5 E:7.0 R:2.0
T-150	AI Training Dataset Provenance and Copyright Protocol	Software Integrity & Provenance	Filter 1 (Regulatory)	L:5.0 T:6.5 E:6.0 R:2.0	L:5.0 T:6.5 E:6.0	L:5.0 T:6.5 E:6.0 R:2.0
T-151	Collaborative R&D IP Escrow and Contribution Tracking	Legal Evidentiary & Notarization	Filter 5 (Incumbent)	L:6.0 T:6.5 E:6.0 R:2.0	L:6.0 T:6.5 E:6.0	L:6.0 T:6.5 E:6.0 R:2.0

ID	Hypothesis Denomination	Thematic Cluster	Elimination Stage	Scores (L/T/E/R)	Dominant Incumbent Alternative	Critical Rejection Factor
T-152	**Zero-Knowledge Attestation for the A2A Agent Economy**	M2M Settlement & Compute	**Filter 4 (Procurement)**	`L:7.0 T:8.5 E:7.5 R:3.5`	`L:7.0 T:8.5 E:7.5 R:3.5`	`L:7.0 T:8.5 E:7.5 R:3.5`
T-153	**Text and Data Mining Remuneration Settlement Protocol**	Legal Evidentiary & Notarization	**Filter 1 (Regulatory)**	`L:4.5 T:6.0 E:5.5 R:2.0`	`L:4.5 T:6.0 E:5.5 R:2.0`	`L:4.5 T:6.0 E:5.5 R:2.0`

End of Technical Report v1.1.1 (EN). Document prepared for formal indexing and public repository dissemination.